

# **VISVESVARAYA TECHNOLOGICAL UNIVERSITY**

**JNANA SANGAMA, BELAGAVI-590018**



**A Project Report On**

## **“BLUE COLLAR(MERN STACK WITH RAZORPAY INTEFRATION)”**

Submitted for partial fulfillment of the requirement for the award of the Bachelor degree in  
Computer Science and Engineering during the year 2025-2026.

**Submitted by**

|                      |   |                   |
|----------------------|---|-------------------|
| <b>SHREYAS K R</b>   | - | <b>4CA22CS096</b> |
| <b>VARINI P</b>      | - | <b>4CA22CS112</b> |
| <b>VARSHITHA H B</b> | - | <b>4CA22CS115</b> |
| <b>YASHAS M</b>      | - | <b>4CA22CS123</b> |

**Under the Guidance of**

**Prof. AKSHATHA T M**

Associate Professor

Dept. Computer Science and Engineering

CIT, Mandya



**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**CAUVERY INSTITUTE OF TECHNOLOGY MANDYA**

**SIDDAIAHNAKOPPALU GATE, SUNDAHALLI, KARNATAKA – 571401**

**2025-2026**

# CAUVERY INSTITUTE OF TECHNOLOGY

SUNDAHALLI, SIDDAlAHNAKOPPALU GATE MANDYA 571401

## DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



### CERTIFICATE

This is to Certify that the project entitled **“BLUE COLLAR (MERN STACK WITH RAZORPAY INTEGRATION)”** is a bonafied work carried out by **SHREYAS K R** bearing USN **4CA22CS096** in partial fulfilment for the degree of 7<sup>th</sup> Semester, Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belgaum during the year **2025- 2026**. It is certified that all corrections or suggestions indicated for Internal Assessment have been incorporated on the report deposited in departmental library. This project report has been approved as it satisfies the academic requirements in respect of the project prescribed for the Bachelor of Engineering Degree.

-----  
**Prof. Akshatha T M**

Associate. Prof Dept. Of  
CSE internal guide  
CIT, MANDYA

-----  
**Prof. Rakshitha B H**

Assistant. Professor &  
HOD Dept. Of CSE  
CIT, MANDYA

-----  
**Dr. Srikantappa A S**

Principal  
CIT, MANDYA

**Name of the Examiners**

1. \_\_\_\_\_

2. \_\_\_\_\_

**Signature with date**

\_\_\_\_\_

\_\_\_\_\_

# CONTENTS

|                                         |                |
|-----------------------------------------|----------------|
| <b>ACKNOWLEDGMENT</b>                   | <b>I</b>       |
| <b>ABSTRACT</b>                         | <b>II</b>      |
| <b>Table of Contents</b>                | <b>Page no</b> |
| <b>CHAPTER 1: INTRODUCTION</b>          | <b>1-4</b>     |
| 1.1 Problem Statement                   | 1              |
| 1.2 Scope and Objective of the Project  | 2              |
| 1.3 Existing System                     | 2-3            |
| 1.4 Proposed System                     | 3              |
| 1.5 Methodology                         | 4              |
| <b>CHAPTER 2: LITERATURE SURVEY</b>     | <b>5-9</b>     |
| 2.1 Paper 1                             | 5-6            |
| 2.2 Paper 2                             | 6-7            |
| 2.3 Paper 3                             | 7-8            |
| 2.4 Paper 4                             | 8-9            |
| <b>CHAPTER 3: SYSTEM SPECIFICATION</b>  | <b>10-11</b>   |
| 3.1 Hardware Requirements               | 10             |
| 3.2 Software Requirements               | 10             |
| 3.3 Functional Requirements             | 11             |
| 3.4 Non - Functional Requirements       | 11             |
| <b>CHAPTER 4: SYSTEM DESIGN</b>         | <b>12-16</b>   |
| 4.1 Architectural Design                | 12             |
| 4.2 Block Diagram                       | 14             |
| 4.3 Sequence Diagram                    | 16             |
| <b>CHAPTER 5: SYSTEM IMPLEMENTATION</b> | <b>17</b>      |
| <b>CHAPTER 6: SYSTEM TESTING</b>        | <b>18-21</b>   |
| 6.1 Test Case Table                     | 19             |
| 6.2 Advantages                          | 20             |
| 6.3 Disadvantages                       | 21             |
| 6.4 Applications                        | 21             |
| <b>CHAPTER 7: RESULTS</b>               | <b>22-27</b>   |
| <b>CONCLUSION</b>                       |                |
| <b>FUTURE ENHANCEMENT</b>               |                |
| <b>REFERENCES</b>                       |                |

## LIST OF FIGURES

| <b>Figures no.</b> | <b>Description</b>    | <b>Page no.</b> |
|--------------------|-----------------------|-----------------|
| Fig:4.1            | System Architecture   | 12              |
| Fig:4.2            | Data Flow Diagram     | 14              |
| Fig:4.3            | The Sequence Diagram  | 16              |
| Fig:7.1            | Worker Sign Up        | 22              |
| Fig:7.2            | Worker Login          | 23              |
| Fig:7.3            | Employer Sign Up      | 24              |
| Fig:7.4            | Employer Login Page   | 25              |
| Fig:7.5            | Our Services          | 26              |
| Fig:7.6            | Live Location Sharing | 27              |

## ACKNOWLEDGMENT

I'm grateful to the management of the **CAUVERY INSTITUTE OF TECHNOLOGY**, for providing us with the opportunity to carry out the project. I have great pleasure in expressing deep sense of gratitude, to our principal **Dr. SRIKANTAPPA A S** for creating an excellent and technically sound environment in our institution.

I'm extremely grateful to **Prof. RAKSHITHA B H, Assistant. Professor & HOD**, Department of Computer Science and Engineering for her valuable suggestions and for helping me in completing my Project.

I would like to express my profound sense of gratitude to our guide **Prof. AKSHATHA T M, Associate Prof** Department of Computer Science and Engineering for the Valuable Suggestions and for helping me in completing our Project.

I'm extremely grateful to my project co-ordinator Prof. **RADHIKA K P, Assistant. Prof**, Department of Computer Science and Engineering for her valuable suggestions and for helping me in completing my Project.

I thank all the teaching and non-teaching staff of Computer Science and Engineering department who has helped me on various occasions during the course of this work.

**SHREYAS K R**

## **ABSTRACT**

The Blue Collar Worker–Client Web Application is designed to digitalize and organize the hiring process for blue-collar workers. In the existing system, workers depend on middlemen, local contacts, and manual communication to find jobs, which leads to delays, miscommunication, and lack of transparency. To overcome these challenges, the proposed system provides a centralized online platform where workers and clients can interact directly.

The application allows workers to create digital profiles, showcase their skills, and search for relevant jobs based on their experience and location. Clients can post job requirements, view worker profiles, shortlist applicants, and manage hiring activities efficiently. The system also includes real-time communication, secure authentication, and structured data management, ensuring a smooth and reliable experience for both parties.

Developed using modern web technologies such as React.js, Node.js, Express.js, and MongoDB, the platform offers fast performance, scalability, and user-friendly interaction. Overall, the project aims to eliminate middlemen, improve transparency, and provide an accessible digital solution to enhance employment opportunities and streamline the hiring process for blue-collar workers.

